

Project Initialization and Planning Phase

Date	30 September 2026
Team ID	SWTID-2026-6111
Project Title	FIT BUDDY AI
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report presents FitBuddy as an AI-powered web application that generates personalized workout plans and nutrition or recovery tips based on user fitness goals. It uses Google Gemini models with FastAPI to provide structured, adaptable fitness guidance.

Project Overview

Objective	To generate personalized 7-day workout plans and goal-aligned nutrition or recovery tips using user information such as age, weight, fitness goal, and preferred workout intensity.
Scope	The project covers AI-based workout generation, nutrition tips, feedback-driven plan updates, user and plan storage with SQLite, and an administrative view of user plans.

Problem Statement

Description	Users need structured, personalized fitness routines and practical nutrition or recovery guidance; the application addresses the difficulty of creating and adapting such plans manually.
Impact	Provides an easy-to-follow, goal-oriented plan and an adaptive feedback loop, helping users refine their routines as preferences or progress change.

Proposed Solution

Approach	Use FastAPI with Google Gemini 1.5 Pro for structured 7-day workout generation and feedback-based updates, and Gemini Flash for fast nutrition or recovery tips; store user and plan data in SQLite.
Key Features	- Personalized 7-day workout plan generation. - Goal-based nutrition and recovery tips. - AI feedback-based workout plan updates. - SQLite storage for users and original/updated plans. - Admin view for registered users and plans.

Proposed Solution Key Features (continued)

	- Feedback-based workout plan updates using Gemini 1.5 Pro. - Fast nutrition/recovery tips using Gemini Flash.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Application and local development environment	System suitable for running the FastAPI application
Memory	Memory for application and local development	Not specified in project document
Software		
Frameworks	Backend framework	FastAPI
Libraries	Templating, ORM/database, API and form support	Jinja2, SQLAlchemy, python-multipart, google-generativeai
Development Environment	Programming/runtime and server tools	Python, pip/virtualenv, Uvicorn, Git
Data		
Application Data	User information and AI-generated workout plans	SQLite database; original and updated plans are preserved